

Scenario Based

Activity

1) Scenario:

CCTV image contains salt and pepper noise

Technique used:

Median Filter

Reason:

Removes salt and pepper noise effectively while preserving image edges.

2) Technique used:

Gaussian Filter

Reason:

Smooths Gaussian noise by averaging nearby pixels and producing a cleaner image.

3) Technique used:

Bilateral Filter

Reason:

Reduces random noise while preserving important edges and fine details

4) Technique used:

Median Filter

Reason:

Eliminates isolated black and white dots without affecting text edges

5) Technique used:

Gaussian Filter

Reason:

Reduces grainy noise when edge preservation is less important

6) Technique used

Histogram equalization

Reason

Improves contrast and makes road details more visible

7) Reason & Technique
contrast stretching

Reason

Expands the intensity range

8) Adaptive Histogram Equalization (CLAHE)

Enhances local contrast and reveals fine internal structures

9) Technique

Histogram equalization

Reason

Increases overall intensity variation, making vehicles and roads easier to distinguish.

10) contrast stretching

Spreads the narrow intensity values across the full grayscale range, improving image contrast